



Hack Away Hunger 2026

dsmHack x Corteva

Submission

Count Me In

Volunteer hours, verified — SNAP kept

80 h

Monthly rule

8

Languages

1 sec

To verify

\$303k

Aid preserved / yr

Live app	https://count-me-in-iowa.netlify.app
Live demo stage	https://count-me-in-iowa.netlify.app/stage
Verify an attestation	https://count-me-in-iowa.netlify.app/verify (no key needed)
Source code (MIT)	https://github.com/chessoren/hackawayhunger
Team	Oren Vidal
Prizes	Grand Prize · Community Impact Award · People's Choice Award

We don't create food. We stop it from disappearing.

Submission at a glance

Volunteering → verified hours → SNAP kept. An SMS & voice assistant in 8 languages helps Iowans meet the new 80-hour rule, and gives pantries the regular volunteers they need.

The problem

Since the July 2025 federal law (P.L. 119-21), adults 18–64 without a child under 14 must prove 80 hours a month of work, training or volunteering to keep SNAP; exemptions for veterans, people experiencing homelessness and former foster youth were removed. Iowa SNAP recipients fell 9% in a year (271,880 → 247,907) while 1 in 8 Iowans face food insecurity. The law already counts volunteering hour for hour — **what's missing is the proof.**

What we built

Person

SMS / voice / web assistant in 8 languages: letter photo → 3-sentence summary + deadline; screening decided by a cited, deterministic rules engine; 80-hour gauge; 3 accessible missions; QR or 4-digit check-in; consent-gated sending; DELETE by SMS.

Host site

10-minute onboarding, QR poster, first name only, one-tap validation signed with Ed25519, grant-ready report, audit log.

State

Monthly PDF attestation with QR; verified in one second; any modification breaks the signature.

Community

Anonymized dashboard ($k \geq 10$), map of opportunity deserts for 99 counties, editable impact model ($\approx$ \$303,000/year for 500 people).

How to test in 2 minutes

1. Open **count-me-in-iowa.netlify.app/stage** and paste a Claude or OpenAI key — it stays in your browser.
2. Text **Hi** (or Hola, Jambo), then tap **Send the sample State letter**.
3. Answer in your own words; pick mission 1, 2 or 3.
4. Demo clock: **Shift starts** → **Scan the QR at the door** → **Shift ends** → scan again → **Confirm**.
5. Text **ATTESTATION**, open it, then visit **/dashboard**.

API keys — temporary.

For the submission, the app runs on each user's own AI key; no key is shipped in the code. **If Count Me In wins, the prize funds the AI, SMS and voice keys so the service runs for everyone.** The demo video was recorded with pre-written AI responses; the live app uses the real APIs. Sites, people and the sample letter are illustrative.



Count Me In

Volunteer hours, verified — SNAP kept

• Live in Iowa

80-hour rule

8 languages

80 h

🕒 Monthly rule

1 sec

🛡️ To verify

3

👥 Winners per hour



Count Me In

(515) 800-8080 · English



Hi

Hi! I'm Count Me In. I help you keep your food assistance (SNAP) under the new 80-hour rule — for free, by text or by phone call.

Private by design: I only keep your first name, phone, language and hours. Nothing goes to the State without your YES. Reply DELETE anytime to erase everything, or HELP to talk to a person. I speak: English · Español · Kiswahili · العربية · Tiếng Việt · Français · Ikinyarwanda.

What's your first name?



Text message



Send the sample State letter



STATE DEPARTMENT OF HUMAN SERVICES
Food Assistance Program - Field Office 77
P.O. Box 0000, Des Moines, IA 50300

SAMPLE
NOTICE

Date of notice: **October 1, 2026**
Case number: XX-XXXXXX

LINDA M.
1400 E Walnut St Apt 2
Des Moines, IA 50316

NOTICE: ABLE-BODIED ADULT WORK REQUIREMENT

Federal law (Public Law 119-21) changed the rules for Food Assistance (SNAP). Our records show you are between 18 and 64 years old and you do not have a child under 14 in your food assistance household.

Beginning with the benefit month of **November 2026**, you must work, volunteer, or take part in a work or training program for at least **80 hours each month**, unless you qualify for an exemption.

WHAT YOU MUST DO: Within **21 days of the date of this notice**, report your hours or ask for an exemption. Return the enclosed form 470-XXXX with proof of hours worked or volunteered.

If you do not meet the requirement, you can receive food assistance for only **3 months in a 36-month period**.

You may be exempt if you are medically certified as physically or mentally unfit for work, pregnant, or caring for an incapacitated person. Volunteer work with a public or nonprofit organization counts toward the 80 hours when it is verified.

Questions? Call the Contact Center at **1-800-000-0000**, Monday to Friday, 8:00 a.m. to 5:00 p.m. You have the right to appeal this decision.

SAMPLE DOCUMENT FOR DEMONSTRATION — Count Me In, Hack Away Hunger 2026. Not an official notice.

Meet Linda

Linda, 58

Des Moines • cleans houses 10 h a week

• Gets SNAP

No car

80 h

📅 Now required

21 days

🕒 To respond

A two-page letter in administrative English. Without help, by spring she is in the line at a pantry that already breaks records.

What changed in 2025

A federal rule just reached hundreds of thousands of lowans



18 → 64

The 80-hour rule now covers adults up to 64



Child under 14

Parents of teenagers are no longer exempt



Exemptions removed

Veterans, homelessness, former foster youth



3 months in 36

Without 80 h or an exemption, SNAP stops

-9%

Iowa SNAP recipients in one year
271,880 → 247,907

1 in 8

Iowans food insecure
73M-meal gap

The real problem

Not willingness. Proof.

01 They don't know if they're covered
_____ or already exempt

02 They can't read the letter
_____ long, English, tight deadline

03 They can't reach hours
_____ no car, irregular shifts

04 They can't prove what they do
_____ volunteering leaves no trace

05 They miss the deadline
_____ one missed report cuts the aid



The other half

Pantries chronically lack **regular weekday volunteers** — and track attendance on paper they can't certify.

SNAP provides ~9 meals for every 1 from food banks. **Every person who loses SNAP lands in the pantry line.**

• The insight • 7 CFR 273.24

The law already counts volunteering. We built the missing proof.



The person

Keeps SNAP — no app, no password

Keeps food aid



The pantry

Gains regular volunteers

Gains volunteers



The State

Gets clean, signed proof

Clean proof

One hour of volunteering serves three people at once.



Count Me In

(515) 800-8080 · English



SNAP work requirement notice —
State Department of Human Services
(sample)

- Starting in November, you must do 80 hours a month of work, volunteering or training to keep your food assistance.
- If you do not, your food assistance can stop after 3 months.
- Volunteering at a nonprofit counts, if your hours are verified.

Deadline: Thu, Oct 22

Report your hours or ask for an exemption by October 22, with the form and proof of your hours.

1-800-000-0000

I set a reminder 3 days before the deadline.

Let's check together if this rule applies to you. What's your first name?



Text message



Send the sample State letter



01

The letter, decoded

Snap a photo. Get 3 sentences.

Text "Hi" to a number — SMS, voice or web. Photograph the State letter, even crooked. Back, in her language: **what it says, the deadline, the one thing to do**, and a reminder 3 days before.

3

Sentences

8

Languages

0

Apps

02

Does this apply to me?

6 questions. A cited rule.

She answers in her own words — *"My knees hurt a bit but I can still work."* The AI turns words into facts; a **deterministic rules engine decides** and cites the regulation. Unsure? A human navigator.

1

Question at a time

0

AI decisions

1 file

Per state



Count Me In

(515) 800-8080 · English



About how many hours a week do you work for pay? (Send 0 if none.)

I clean houses, about 10 hours a week

The 80-hour rule applies to you. Don't worry: work, training and volunteering all count, in any mix. I will help you get there.

Rule cited

7 CFR 273.24 — Time limit for able-bodied adults (work = paid, in-kind, unpaid or volunteer work)

Rules file IA 2026.10-hackathon

Your paid work gives about 43 hours a month. Do you have training or English class hours this month? Send the number (0 if none).

0



Text message



Send the sample State letter





Count Me In

(515) 800-8080 · English



example: "weekday mornings" or
"Tuesdays after 2pm".

I am free anytime

I clean houses Monday and Wednesday
mornings

Here are missions that fit you:

1 Food distribution — Grace Church
Community Grocery

Tue, Oct 13, 09:00–13:00 (4 h) · 12 min
walk

2 Food distribution — Eastside
Neighborhood Pantry

Sat, Oct 10, 08:00–12:00 (4 h) · 19 min
walk

3 Welcome desk — Downtown
Community Kitchen

Sun, Oct 11, 12:00–15:00 (3 h) · Route 4
— E 14th St, about 33 min · seated work
Reply 1, 2 or 3.

1

2

3



Text message



Send the sample State letter



03

Three missions, never more

Ranked by real access.

A 12-minute walk or a DART bus line, around her cleaning schedule, optimized to reach **80 hours before the deadline**. She replies "1". At the door: scan the QR, or text a 4-digit code from any basic phone.

3

✓ Choices max

12 min

🚶 Walk

4 digits

☎ Basic phones

The paper sign-in sheet, replaced

The screenshot shows the 'Host site' app interface. At the top, it says 'Host site' and 'The paper sign-in sheet, replaced'. Below this, there are two tabs: 'One-tap validation' (selected) and 'First name only'. A yellow button '+ Register a host site' is in the top right. The main content area shows 'Coordinator - Pastor Dave' and a dropdown menu for 'Grace Church Community Grocery'. Below this, it says 'Tuesday, October 13' and '1 expected'. A green card shows a volunteer named 'Linda' with a slot 'L' for 'Food distribution - 09:00-13:00' and a status '4 h confirmed'. At the bottom, there is a note: 'You only see a first name and a slot — never why someone volunteers, never their SNAP file. Confirmations are signed with this site's key (8E32-6827-81A9-6E12).' and a button 'Request more volunteers'.



10 min
to become a host site



One tap
confirmation, Ed25519 signed



First name
never the SNAP file



Grant-ready
monthly report & CSV

Verified in one second. Forged in none.



Verify an attestation

For caseworkers · one second · nothing stored

• No account • No AI key

Authentic

Not modified since it was signed by Count Me In.

Linda 2026-10 47 / 80 h

Name Month Total hours

Date	Organization	Mission	In-Out	Hours	Validated by
2026-10-13	Grace Church Community Grocery	Food distribution	09:02-12:59	4 h	Pastor Dave key B:32-6827-81A9-6E12
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43 h	—

Volunteering 4 h · Training 0 h · Paid work declared by the person 43 h · ID CMI-202610-02606C · issued 10/13/2026, 12:59:03 PM · rules 1A 202610-hackathon · platform key E/26-FAB0-9CC3-7F5E

Download the PDF

Verify another document

• **Authentic** signed by the platform key



Verify an attestation

For caseworkers · one second · nothing stored

• No account • No AI key

Not valid

Signature does not match the document. It was modified after signing, or not signed by Count Me In.

Linda 2026-10 51 / 80 h

Name Month Total hours

Date	Organization	Mission	In-Out	Hours	Validated by
2026-10-13	Grace Church Community Grocery	Food distribution	09:02-12:59	8 h	Pastor Dave key B:32-6827-81A9-6E12
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43 h	—

Volunteering 8 h · Training 0 h · Paid work declared by the person 43 h · ID CMI-202610-02606C · issued 10/13/2026, 12:59:03 PM · rules 1A 202610-hackathon · platform key E/26-FAB0-9CC3-7F5E

Verify another document

The attestation is signed with the Count Me In platform E825519 key 1E/26-FAB0-9CC3-7F5E. Each line references a coordinator-signed ledger entry. Changing any

+4 hours forged signature breaks instantly



Impact & opportunity deserts

lowa · 99 counties · anonymized, k ≥ 10

<10

People supported (live)

<10

Recognized as exempt

<10

Covered, on a plan

4

Volunteer hours verified

Where 80 hours are out of reach without a car

Opportunity desert · 64

Tight · 18

Enough hours · 17

64

Deserts

64 of 99 counties

are opportunity deserts: car-free accessible volunteer hours cover less than half of what people subject to the rule need. Tap a county.

County data (CSV)

Who uses this map



Food banks & pantries

Where to open the next host site



Funders & companies

Measurable impact in meals and dollars



Officials & Iowa HHS

Counties under strain, failure reasons



Researchers & press

Aggregated, downloadable data

Model: SNAP participants = population × Iowa rate (247,907 recipients, Iowa HHS May 2026); subject = 14% of participants; 35% need ~20 volunteer h/month. Supply: ~1 volunteer-hosting organization per 1,200 residents × 150 h/month × car-free share (60% with fixed-route transit, 40% mid-size, 25% small rural counties), plus registered host sites. The pilot replaces the baseline with the live site registry and transit data.

06 · For Iowa

64

of 99 counties

are **opportunity deserts**: car-free volunteer hours cover less than half of what people under the rule need.

Neutral data — where to open the next host site, or where to ask for a waiver. Never a cell under 10 people.

Food aid preserved per year

\$303,318

for 500 people supported · every number editable on the live dashboard

$500 \times 40\% \text{ would lose SNAP} \times 75\% \text{ kept} = \mathbf{150 \text{ people}}$

$150 \times \$168.51 \text{ Iowa average benefit} \times 12 \text{ months}$

89,211

meals / year

36,000

volunteer hours / year

~\$5k

running cost / year

How it's built

AI understands. Rules decide. A human settles doubts.

The person

SMS · voice · web

The coordinator

QR · one-tap validation

Public & officials

Dashboard · map

Multilingual AI agent

Claude / OpenAI · structured outputs

Letter reader

Vision → summary + deadline

Rules engine

One YAML per state, cited

Matching

Walk · DART · schedule

Signed hours ledger

Hash chain · Ed25519 · audit

Human relay

Navigator console

Verifiable attestation

PDF + QR

Iowa HHS

Only after a YES

Anonymized data

$k \geq 10$ by county



Count Me In

(515) 800-8080 · Kiswahili



Jambo

Habari! Mimi ni Count Me In. Ninakusaidia kubaki na msaada wako wa chakula (SNAP) chini ya sheria mpya ya saa 80 — bure, kwa ujumbe au kwa simu.

Faragha kwanza: ninahifadhi jina lako la kwanza, simu, lugha na saa tu. Hakuna kinachoenda kwa Serikali bila YES (NDIYO) yako. Jibu DELETE (FUTA) wakati wowote kufuta kila kitu, au HELP (MSAADA) kuongea na mtu.
Ninazungumza: English · Español · Kiswahili · العربية · Tiếng Việt · Français · Ikinyarwanda.

Jina lako la kwanza ni nani?



Text message



Send the sample State letter



Dignity · access · privacy

Any phone. Any language. Always a human.



8 languages + voice for people who read little



Any phone keywords & 4-digit codes work without AI



HELP, always a human navigator one word away



DELETE by SMS minimal data, pseudonymous ledger



One option never a condition to receive food

What's next

Oct 2026

MVP

Full SMS journey, QR, attestation, map

Nov 26 – Mar 27

DMARC pilot

3–5 host sites · English + Spanish

2027

All of Iowa

55 counties · 8 languages · HHS link

2028

Other states

One rules file per state · open source

We don't create food. **We stop it** from disappearing.



count-me-in-iowa.netlify.app

>>

Judging runs on your own AI key (temporary). If we win, the prize funds the keys for everyone.

Inspiration

Linda is 58, lives in Des Moines, cleans houses ten hours a week and has no car. In October a letter tells her that, under the 2025 federal law, she must now show **80 hours a month** of work, training or volunteering, or lose her SNAP food assistance. She has 21 days.

Linda is a composite, but her situation is real for many Iowans. The law (P.L. 119-21) raised the SNAP work-rule age limit from 54 to 64, limited the parent exemption to households with a child under 14, and removed the exemptions for veterans, people experiencing homelessness and former foster youth. Iowa SNAP recipients fell **9% in a year** (271,880 → 247,907, May 2025 → May 2026), while **1 in 8 Iowans** face food insecurity. Everyone who loses SNAP ends up in a pantry line that is already breaking records.

Then we found the detail that shaped the project: **federal rules already count volunteering toward the 80 hours, hour for hour (7 CFR 273.24), as long as the hours are verified.** People don't lose SNAP because they refuse to work. They lose it because they can't prove what they do. Meanwhile pantries are short of regular weekday volunteers and track attendance on paper. So one well-recorded hour serves three people at once: **the person keeps their food aid, the pantry gains a volunteer, and the State gets clean proof.**

What it does

Count Me In turns volunteering into verified hours.

For the person — any phone, any of 8 languages. No app, no account: they text a number (or call, or use the web).

- **Letter decoded:** they photograph the State letter and get back three plain sentences, the deadline and the one thing to do, with an automatic reminder.
- **Screening:** six questions, one at a time, answered in their own words. A deterministic rules engine — not the AI — decides "probably exempt", "covered" or "let's check with a navigator", and cites the regulation.
- **Planning:** an 80-hour gauge, then at most **three missions** ranked by real access (walking time, DART bus, work schedule, language, physical limits).
- **Logging:** check-in by **QR code** or a **4-digit code** texted from any basic phone.
- **Always in control:** HELP reaches a human, DELETE erases everything.

For host sites. Pantries, churches, gardens and corporate volunteer programs onboard in about 10 minutes and print a QR poster. The coordinator sees only a first name and a time slot, never the SNAP file. They confirm hours in one tap, and each confirmation is signed with the site's Ed25519 key. They get a monthly report ready for grant applications.

For the State. Each month the person gets an attestation as a PDF with a QR code. A caseworker scans it and knows in one second that it is authentic; **change a single digit and the signature breaks.** Nothing is sent to Iowa HHS without the person's explicit YES, and that consent is logged.

For the community. A public, anonymized dashboard (no cell under 10 people) and a **map of opportunity deserts**: for each of Iowa's 99 counties, how many volunteer hours are reachable without a car compared with the hours needed. Under our model, 64 counties, mostly rural, fall short.

Impact you can check in your head: 500 people supported $\times$ 40% who would lose SNAP $\times$ 75% who keep it with the tool = 150 people. $150 \times \$168.51$ (Iowa average benefit, May 2026) $\times$ 12 $\approx$ **\$303,000 of food aid preserved per year**, plus about 36,000 volunteer hours for host sites. All of this runs on a few thousand dollars a year.

■ How we built it

- **React + TypeScript + Vite**, mobile-first and accessible: large touch targets, high contrast, right-to-left Arabic, a Burmese font, and a voice mode using the Web Speech API.
- **One channel-agnostic agent** — the same code answers the web phone and the **Twilio** SMS and Voice webhooks. Keywords and site codes are handled first, without AI, so basic phones and outages still work. AI is used only for free text.
- **AI layer:** Claude (Anthropic SDK) or OpenAI, always with **structured JSON outputs**. It does three things: understand free text, read letter photos with vision and OCR, and translate.
- **Rules engine:** `rules/iowa.yaml`, one readable file per state with its sources. When the law changes, you edit the file, not the code.
- **Signed ledger:** append-only and hash-chained, with Ed25519 signatures (TweetNaCl). It stores a salted pseudonym, never a phone number.
- **Attestations:** signed server-side, built with **pdf-lib**, a QR code, and a verification page that also works by camera scan (**jsQR**).
- **Map:** a custom SVG renderer of Iowa county GeoJSON — no third-party map tiles.
- **Hosting:** serverless functions on **Netlify**.
- **Quality: Vitest** (20 tests, including forgery and tampering cases) and a full **Playwright** end-to-end run.

API keys — temporary. For the judging, the app asks each user for their own Claude or OpenAI key. The key stays in the browser and is never in the code. **If we win, the prize will fund the AI, SMS and voice keys so the service runs for everyone**; people like Linda will never see a key. Because we shipped no keys, the demo video was recorded with pre-written AI responses; the live app uses the real APIs.

■ Challenges we ran into

- **Getting the law exactly right.** Sources contradicted each other: early drafts set the parent threshold at a child under 7. We encoded only rules we could cite.
- **Keeping the AI out of decisions without sounding robotic.** The AI can only fill in facts; the rules engine decides. Making that still feel human took several iterations.
- **Translating without breaking SMS.** Option numbers, codes, times and keywords must survive translation intact.
- **Making proof genuinely hard to fake.** We wrote tests that try to forge an attestation, for example turning 4 hours into 8, and made sure they fail.
- **Staying honest.** We used Iowa's real average benefit (\$168.51) rather than a rounder figure, even though it lowers our headline number.

■ Accomplishments that we're proud of

A complete journey you can test in two minutes: from "Hi" to a signed, verifiable attestation. A rules file a paralegal can read. Proof a caseworker can trust in one second. It works on any phone, in eight languages, with a human always one word away, and privacy is built in from the start.

■ What we learned

The bottleneck is **verification, not willingness**. "AI-first" doesn't have to mean "AI decides": a cited rules engine makes the tool easier to trust and to explain. And dignity has to be designed in. That means calling them "missions" and "hours that count", making Count Me In volunteers indistinguishable from everyone else, and sending nothing without a yes.

■ What's next for Count Me In

1. **Pilot in the DMARC network** (Nov 2026 – Mar 2027): 3–5 host sites, English and Spanish.
2. **Validate the attestation format with Iowa HHS.**
3. **Scale to Food Bank of Iowa's 55 counties**, then other states with one rules file each.

Next steps include moving the ledger to server-side PostgreSQL, loading DART's full transit feed, and funding the keys. The code is open source (MIT), with a nonprofit host and dsmHack volunteers for maintenance. We'd love Corteva to be the first corporate host site.

We don't create food. We stop it from disappearing.

Demo host sites, people and the sample letter are illustrative; no partnership is implied.